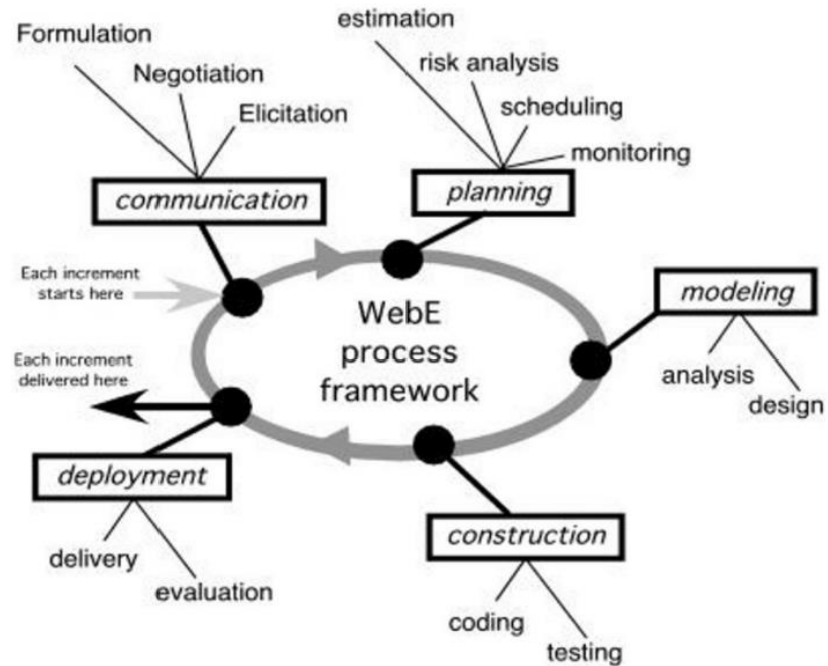
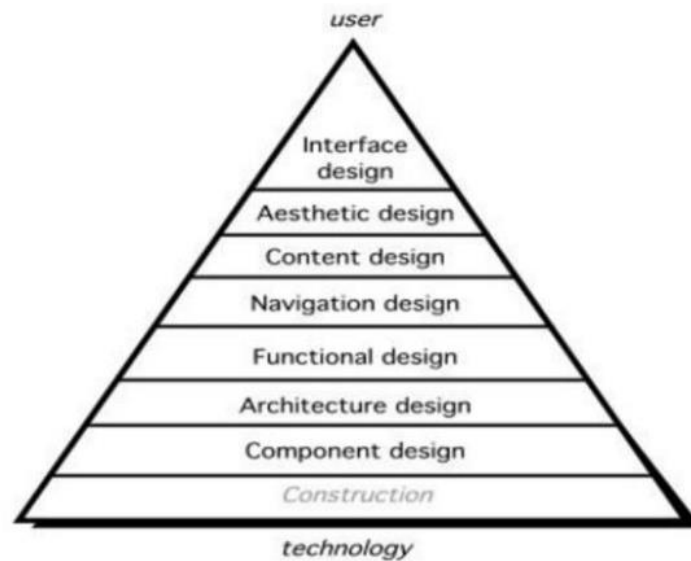


المقالي المحتمل علي حسب كلام الدكتور في اخر محاضره

WebE Process Activities & Actions

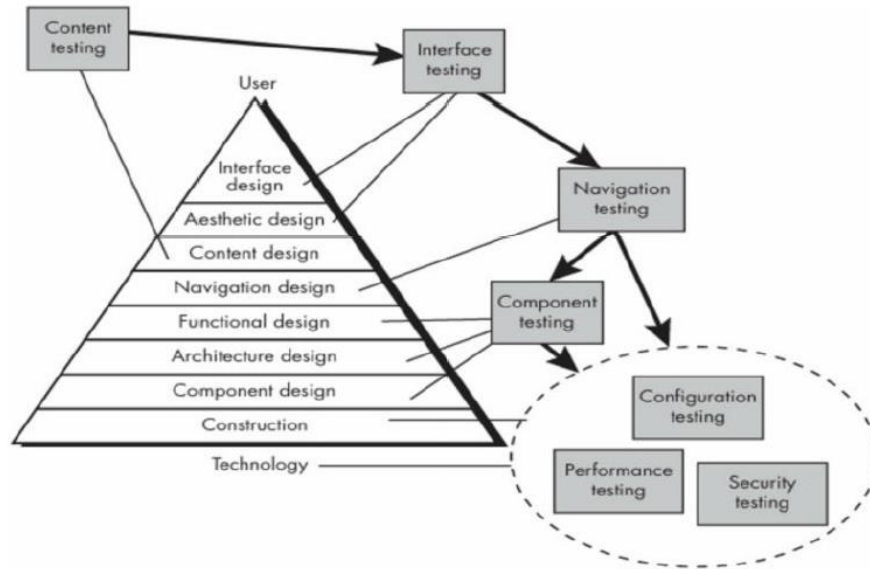


Design Actions



The Testing Process

Note that: In the following figure, as the testing flow proceeds from left to right and top to bottom, user-visible elements of the WebApp design (top elements of the pyramid) are tested first, followed by infrastructure design elements.



صل علي سيدنا محمد

ادعوا لي بالتوفيق ولكم بالمثل

افتكروا ابي و جدتي في دعائكم

Estimation (Time and Effort)

Estimation focus for most WebE projects is on macroscopic, rather than microscopic, issues. The WebE team assesses whether a planned WebApp increment can be developed with available resources according to defined schedule constraints.

■ **Two** viable (and quick) **approaches** to detailed estimation:

1. *Usage scenario-based estimation*

- Examines the usage scenarios (e.g., use cases) for the increment, compares them to previous data on average effort (E_{avg}) for previous increments. E_{avg} is the average effort (in person-days) required to deploy a usage scenario.
- Adjust estimates based on perceived complexity
- Ex: The following Figure for the fourth SafeHomeAssured.com increment, where the history for the team indicates that E_{avg} is 14 person-days. The two usage scenarios have complexity multipliers of 2.5 and 2.0, respectively. Hence, the overall effort required to implement the fourth increment is estimated to be 63 person-days. That is, if 63 person days are required to model, construct, and deploy an increment, the work could be accomplished by **one person working for**

63 days, two people working for 32 days, three people working for 21 days, and so forth. Sadly, **a linear relationship does not exist**. In reality, as more people become involved in a project, more effort is spent on communicating and coordinating (meetings, e-mail, etc.). Additional time is spent on things that have nothing to do with the project (e.g., telephone calls, coordinating the softball league), administrative issues (e.g., applying for a new health insurance plan), and other “nonproductive” work. Consequently, it may take three people 25 or more calendar days to achieve 63 person-days of effort.

Usage Scenario	E_{avg}	Complexity	Effort
Develop a layout for the space to be monitored	14	2.5	35
Get recommendations for sensor layout for my space	14	2.0	28
Totals			63

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- **Stress testing** is a continuation of load testing, but in this instance the variables, N , T , and D are forced to meet and then exceed operational limits.
 - **N**: the number of concurrent users.
 - **T**: the number of online transactions per unit of time.
 - **D**: the data load processed by the server per transaction.
 - Load testing can also be used to assess recommended connection speeds for users of the WebApp.
 - **Overall throughput (P)** is computed in the following manner: $P = N * T * D$.
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-
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- **Ex.:** consider a popular sports news site. At any given time, 4000 concurrent users submit a request (a transaction T) once every 30 seconds on average. Each transaction requires the WebApp to download a news article that averages 12 Kbytes in length.
 - Therefore, $N = 4000$ users, $T = 0.033$ transactions per second per user, $D = 12$ Kbyte per transaction.
 - Throughput can be calculated as: $P = 4000 * 0.033 * 12 \approx 1600$ Kbyte/s
 - The network connection for the server would therefore have to support this average data rate and should be tested to ensure that it does.

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